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cludes encryption and cryptographic authentication, which impose major challenges in terms of the cost of power and bandwidth.

Performance. Performance encompasses bandwidth, data rates, latency and packet prioritisation (QoS) techniques. In fact, transmission time has a major impact on performance.

Secondary design challenges

Other factors that need to be considered when developing and deploying any WSN solution are cost-effectiveness, flexibility and scalability, fault tolerance, redundancy and robustness, self-organising and self-healing, routing, energy-efficiency constraints, transmission time, interference and environment, and blocked paths.

Cost-effectiveness. With the advances in semiconductor technology, WSN embedded products have become cheaper, especially for single-chip solutions that comprise microcontrollers, transceivers and on-chip memory. A key factor in the cost of wireless embedded processors is the amount of memory required. Developers of sensor networks will expect access to a range of chips or wireless microprocessors/microcontrollers with optimised memory footprints to meet the needs of a range of applications.

Flexibility and scalability. WSN architecture must be flexible and adaptive enough to deploy in a wide range of usage scenarios. The number of sensors in different deployments may vary from thousands to millions. This will have a significant impact on network maintenance and diagnostics. Therefore network management must be flexible enough to sustain this huge network of sensors. Scalability—primarily governed by access to media—is

another challenge in a WSN.

Fault tolerance, redundancy and robustness. Node loss is an important issue to consider in a WSN. Since sensors can be deployed everywhere, e.g., on bridges, industrial sites, homes, farms, etc, these will be subjected to harsh environmental conditions and may be vandalised, stolen or damaged. Additionally, power surges, blackouts or brownouts can cause nodes to fail. So it's important to configure and deploy a network that can isolate individual points of failure, eliminate or mitigate their impact, and continue to operate as a whole, maintaining very high end-to-end fault tolerance and robustness against nodes failure.

Self-organising and self-healing. A WSN should provide extremely high reliability and predictability for years on end by applying self-organising and self-healing intelligence to continuously adapt to unpredictable conditions. There are some applications where sensors are deployed sporadically in a geographical location and these have to organise themselves and find out where they are in relation to each other. The exact structure of a WSN cannot be predetermined in most situations due to variable dependencies such as location of devices and radio propagation during network formation. In addition to joining a network, devices could leave, be forced to leave or potentially re-associate elsewhere within the network—all of which could cause a change in network structure.

Routing. In addition to managing network joining, de-association and re-association, the network formation process requires various routing methodologies. Traditional routing schemes are not designed to be energy- or processing-efficient and are of little use in a sensor network. In that

case, it is best to keep the routing to a minimum, and activate only when absolutely necessary.

Energy-efficiency constraints. Sensor nodes are typically battery-driven. These are too small and too numerous to replace or recharge batteries. Moreover, microsensor networks are often deployed in remote or dangerous environments. Hence, energy-efficiency is a very serious concern—the increase in the lifespan of sensor nodes is one of the main design and implementation challenges. Very rarely is it possible to plug a device into an electrical power outlet.

Transmission time. An issue sometimes overlooked is the length of time it takes for transmission of data packets. In fact, transmission time has a major influence on performance, quality, power consumption and interference. So reducing it is a major design challenge.

Interference and environment. In some deployments, interference from different wireless technologies may need to be addressed. Heavy network traffic from other wireless networks sometimes imposes restrictive availability of the RF channel due to a constant state of interference. The type of building also dramatically affects the RF environment. Steel-reinforced concrete floors, stone walls and similar construction materials, for instance, introduce high levels of attenuation and multi-path fading.

Blocked paths. Sometimes, changes in the physical environment block communication links. Wireless paths are established between devices based on the immediate RF environment and available neighbours when a network is first deployed. These variables often change; paths may later be blocked by new equipment, or very small changes may take place in device position.

Design considerations

The important design considerations when building an extremely reliable and predictable WSN are:

1. What will the system look like?
2. What will the network topology